

MEDICAL MEDICINE ACCORDING TO DISEASE

HANDBOOK

Comprehensive Clinical Reference Guide

Sixth Edition | 2026

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CHAPTER 1: CARDIOVASCULAR DISEASES

1.1 HYPERTENSION

Definition:

Hypertension is a chronic medical condition characterized by persistently elevated arterial blood pressure. Defined as systolic BP ≥ 140 mmHg and/or diastolic BP ≥ 90 mmHg on two separate readings.

Causes:

- Primary (Essential): 95% of cases - no identifiable cause
- Secondary: Renal disease, endocrine disorders, medications, aortic coarctation

Symptoms:

- Often asymptomatic (the "silent killer")
- Headaches, dizziness, blurred vision, epistaxis (in severe cases)
- Symptoms of end-organ damage: chest pain, shortness of breath, confusion

Diagnosis:

- Blood pressure measurement (office, ambulatory, home)
- Lab: Complete blood count, serum creatinine, electrolytes, lipid panel, glucose, urinalysis
- ECG, echocardiogram for target organ damage

Pharmacological Treatment:

Drug Class	Examples	Mechanism	Common Side Effects
ACE Inhibitors	Lisinopril, Enalapril, Ramipril	Blocks ACE; reduces angiotensin II	Cough, hyperkalemia, angioedema
ARBs	Losartan, Valsartan, Candesartan	Blocks AT1 receptors	Same as ACE but less cough

Beta Blockers	Metoprolol, Atenolol, Propranolol	Blocks beta-adrenergic receptors	Bradycardia, fatigue, bronchospasm
Calcium Channel Blockers	Amlodipine, Nifedipine, Verapamil	Blocks calcium channels	Edema, flushing, constipation
Diuretics	Hydrochlorothiazid e, Furosemide, Spironolactone	Promotes sodium and water excretion	Electrolyte imbalance, hypokalemia
Alpha Blockers	Doxazosin, Prazosin	Blocks alpha-1 receptors	Postural hypotension

Non-Pharmacological Management:

- DASH diet (high fruits, vegetables, low sodium)
- Exercise: 150 minutes moderate/week
- Weight loss: 5-10% BMI reduction
- Limit alcohol: ≤ 1 drink/day women, ≤ 2 men
- Sodium intake $< 2,300$ mg/day

Guidelines:

- JNC 8 Guidelines: For >60 years, target BP $< 150/90$; for <60 , target $< 140/90$
- AHA 2022: For all, target BP $< 130/80$

1.2 CORONARY ARTERY DISEASE (CAD)

Definition:

Atherosclerotic plaque buildup within coronary arteries, causing myocardial ischemia.

Risk Factors:

- Modifiable: Smoking, hypertension, diabetes, hyperlipidemia, obesity
- Non-modifiable: Age, male gender, family history

Clinical Presentation:

- Stable angina: Chest pain on exertion, relieved with rest
- Unstable angina: Chest pain at rest, increasing frequency
- Myocardial infarction: Prolonged chest pain, NSTEMI or STEMI

Diagnosis:

- ECG: ST changes, pathological Q waves
- Cardiac enzymes: Troponin I/T (elevated in MI)
- Stress test: Exercise tolerance, nuclear imaging
- Coronary angiography: Gold standard for stenosis

Pharmacological Treatment:

A. Anti-ischemic Therapy:

- Beta-blockers: Metoprolol, Bisoprolol
- Calcium channel blockers: Diltiazem, Verapamil (for vasospastic)
- Long-acting nitrates: Isosorbide mononitrate
- Ranolazine (for refractory angina)

B. Antiplatelet Therapy:

- Aspirin 81-325 mg daily
- P2Y12 inhibitors: Clopidogrel, Ticagrelor, Prasugrel (after PCI)

C. Lipid-Lowering Therapy:

- High-intensity statins: Atorvastatin 80 mg, Rosuvastatin 40 mg
- Ezetimibe (add-on therapy)
- PCSK9 inhibitors: Alirocumab, Evolocumab (for familial hyperlipidemia)

D. Anticoagulation:

- Heparin (for ACS)
- Enoxaparin (LMWH)

Surgical Interventions:

- Percutaneous coronary intervention (PCI) with stent
- Coronary artery bypass grafting (CABG)

Secondary Prevention:

- Lifestyle modifications
 - ACE inhibitors or ARBs
 - Spironolactone (for post-MI)
 - Glycemic control in diabetics
 - Smoking cessation
 - Cardiac rehabilitation program
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1.3 HEART FAILURE (HF)

Definition:

Impaired cardiac pumping ability to meet metabolic demands of the body.

Classification:

- HFrEF: $EF \leq 40\%$ (Heart failure with reduced ejection fraction)
- HFpEF: $EF \geq 50\%$ (Heart failure with preserved ejection fraction)
- HFmrEF: EF 41-49%

Stages:

- Stage A: At risk for HF but without structural heart disease
- Stage B: Structural heart disease but without symptoms
- Stage C: Structural disease with current or previous symptoms
- Stage D: Refractory HF requiring advanced interventions

Symptoms:

- Dyspnea on exertion, orthopnea, paroxysmal nocturnal dyspnea
- Fatigue, weakness
- Edema, ascites, hepatomegaly

Diagnosis:

- BNP/NT-proBNP: Elevated
- Echocardiogram: EF, valvular disease
- Chest X-ray: Cardiomegaly, pulmonary edema

Pharmacological Treatment for HFrEF:

Drug	Indication	Mechanism	Monitoring
ACE Inhibitors/ARBs	All HFrEF	Reduce afterload, neurohormonal modulation	Renal function, potassium
Beta-Blockers	Stable HFrEF	Decrease HR, mortality benefit	HR, BP, signs of decompensation
ARNI (Sacubitril/Valsartan)	HFrEF, NYHA II-III	Combines ARB and neprilysin inhibitor	Renal function, BP
Mineralocorticoid Antagonists	HFrEF (NYHA II-IV)	Block aldosterone, reduce mortality	Potassium levels
SGLT2 Inhibitors	HFrEF (NYHA II-IV)	Empagliflozin, Dapagliflozin	Blood glucose, volume status
Diuretics	Symptomatic congestion	Reduce preload and volume	Weight, electrolytes

Management:

- Sodium restriction (< 2g/day)
- Fluid restriction (1.5-2L/day)
- Daily weight monitoring (2-3 lb/day weight gain indicates worsening)

- Cardiac resynchronization therapy (CRT) for QRS $\geq$ 150ms
- ICD for EF $\leq$ 35% with symptoms
- Ventricular assist devices (VAD) or transplant for end-stage

1.4 DYSLIPIDEMIA

Definition:

Abnormal lipid profiles - elevated total cholesterol, LDL, triglycerides, or low HDL.

Type	Total Cholesterol	LDL	HDL	Triglycerides
Normal	< 200 mg/dL	< 100 mg/dL	> 60 mg/dL (M), > 50 mg/dL (F)	< 150 mg/dL
Moderate	200-239 mg/dL	100-159 mg/dL	40-59 mg/dL	150-199 mg/dL
High	$\geq$ 240 mg/dL	$\geq$ 160 mg/dL	< 40 mg/dL	$\geq$ 200 mg/dL

Treatment Targets (AHA/ACC 2023):

- Very high risk: LDL < 55 mg/dL
- High risk: LDL < 70 mg/dL
- Moderate risk: LDL < 100 mg/dL

Pharmacological Treatment:

Drug Class	Examples	Effect	Indication
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Statins	Atorvastatin, Rosuvastatin	↓ LDL 30-55%	First-line for all elevated LDL
Ezetimibe	Ezetimibe	↓ LDL 15-20%	Add-on or statin intolerant
PCSK9 Inhibitors	Alirocumab, Evolocumab	↓ LDL up to 60%	Familial hyperlipidemia, statin failure
Fibrates	Fenofibrate, Gemfibrozil	↓ TG 30-50%	Hypertriglyceridemia
Niacin	Niacin	↓ TG, ↑ HDL	Mixed dyslipidemia
Bempedoic Acid	Bempedoic Acid	↓ LDL 15-20%	Statin-intolerant

CHAPTER 2: RESPIRATORY DISORDERS

2.1 ASTHMA

Definition:

Chronic inflammatory airway disease characterized by reversible airway obstruction, bronchial hyperresponsiveness.

Classification:

- Intermittent: Symptoms ≤ 2 days/week, nighttime ≤ 2 /month
- Mild Persistent: Symptoms > 2 /week but daily
- Moderate Persistent: Daily symptoms affecting activity
- Severe Persistent: Continuous symptoms, frequent exacerbations

Triggers:

- Allergens: Pollen, dust mites, animal dander
- Irritants: Smoke, pollution, chemicals
- Exercise, cold air
- Respiratory infections

Pharmacological Treatment:

A. Quick-Relief Medications (Rescue):

- SABA: Albuterol, Levalbuterol, Terbutaline
- Anticholinergics: Ipratropium (for severe exacerbations)

B. Long-Term Control Medications:

- Inhaled Corticosteroids (ICS): Fluticasone, Budesonide, Beclomethasone
- LABA: Salmeterol, Formoterol (used only with ICS)
- Combination Inhalers: Fluticasone/Salmeterol, Budesonide/Formoterol
- Leukotriene Modifiers: Montelukast, Zafirlukast
- Biologics: Omalizumab (anti-IgE), Mepolizumab (anti-IL-5), Dupilumab
- Theophylline: For severe refractory asthma

Stepwise Therapy (GINA Guidelines):

Step	Therapy
Step 1	SABA as needed
Step 2	Low-dose ICS + SABA as needed
Step 3	Low-dose ICS/LABA + SABA as needed
Step 4	Medium-dose ICS/LABA + SABA as needed
Step 5	High-dose ICS/LABA + Add-on therapy (tiotropium, biologics)

2.2 CHRONIC OBSTRUCTIVE PULMONARY DISEASE (COPD)

Definition:

Progressive airflow limitation, not fully reversible, characterized by chronic bronchitis and emphysema.

Causes:

- Smoking (primary cause)
- Occupational exposure (dust, chemicals)
- Alpha-1 antitrypsin deficiency (genetic)

Clinical Presentation:

- Chronic cough with sputum production (most mornings)
- Dyspnea (progressive, worse with exertion)
- Barrel chest, hyperinflation
- Cyanosis, peripheral edema (later stages)

Diagnosis:

- Spirometry ($FEV_1/FVC < 0.70$ post-bronchodilator)
- Chest X-ray: Hyperinflation, flattening of diaphragm
- ABG: Hypoxemia, hypercapnia (late)

Pharmacological Treatment:

A. Bronchodilators:

- Short-acting: SABA (Albuterol), SAMA (Ipratropium)
- Long-acting: LABA (Salmeterol), LAMA (Tiotropium), LAMA/LABA combinations

B. Inhaled Corticosteroids:

- For frequent exacerbations ($\geq 2/\text{year}$), eosinophilia $\geq 300/\mu\text{L}$
- Triple therapy: LAMA/LABA/ICS combination

C. Phosphodiesterase-4 Inhibitors:

- Roflumilast (for severe COPD with chronic bronchitis)

D. Theophylline: Third-line

E. Pulmonary Rehabilitation:

- Exercise training, education, nutrition counseling

Oxygen Therapy:

- Home oxygen if $\text{SpO}_2 < 88\%$ or $\text{PaO}_2 < 55 \text{ mmHg}$
 - 2-4 L/min to maintain $\text{SpO}_2 > 90\%$
 - Travel oxygen for long-distance
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2.3 PNEUMONIA

Definition:

Infection of pulmonary parenchyma causing inflammation and consolidation.

Types:

- Community-acquired pneumonia (CAP)
- Hospital-acquired pneumonia (HAP)
- Ventilator-associated pneumonia (VAP)
- Aspiration pneumonia

Pathogens:

Type	Common Organisms
CAP	<i>S. pneumoniae</i> , <i>H. influenzae</i> , <i>M. pneumoniae</i> , <i>C. pneumoniae</i> , <i>Legionella</i>
HAP/VAP	<i>S. aureus</i> (MRSA), <i>P. aeruginosa</i> , <i>Klebsiella</i> , <i>E. coli</i>
Atypical	<i>Mycoplasma</i> , <i>Chlamydia</i> , <i>Legionella</i>

Treatment:

CAP (Outpatient):

- Macrolide: Azithromycin 500 mg x 1 then 250 mg x 4 days
- OR Doxycycline 100 mg BID
- OR Respiratory fluoroquinolone: Levofloxacin 750 mg daily

CAP (Inpatient - Non-severe):

- Ceftriaxone 1-2g daily + Azithromycin 500 mg daily
- OR Levofloxacin 750 mg daily monotherapy

CAP (Inpatient - Severe):

- Piperacillin/Tazobactam 4.5g Q6H + Vancomycin
- OR Cefepime 2g Q8H + Vancomycin
- Add atypical coverage (Azithromycin or Levofloxacin)

HAP/VAP:

- Empiric: Broad-spectrum (anti-pseudomonal beta-lactam + vancomycin)
- De-escalate based on culture results

CHAPTER 3: ENDOCRINE & METABOLIC DISEASES

3.1 DIABETES MELLITUS

Definition:

Chronic metabolic disorder characterized by hyperglycemia due to absolute or relative insulin deficiency.

Classification:

Type	Characteristics	Treatment
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Type 1	Autoimmune destruction of beta-cells; insulin-dependent	Insulin
Type 2	Insulin resistance with relative deficiency	Lifestyle + oral agents + insulin
Gestational	Onset during pregnancy	Insulin, Metformin
LADA	Type 1 presenting in adulthood	Insulin early

Diagnostic Criteria:

- Fasting glucose ≥ 126 mg/dL
- 2-hour OGTT ≥ 200 mg/dL
- HbA1c $\geq 6.5\%$
- Random glucose ≥ 200 mg/dL with symptoms

Pharmacological Treatment for Type 2 Diabetes:

A. First-Line:

- Metformin: 500-2000 mg/day; GI side effects, contraindicated in renal impairment (eGFR < 30)

B. Second-Line Additions:

Drug Class	Examples	Mechanism	Side Effects
Sulfonylureas	Glipizide, Glimepiride	$\uparrow$ Insulin secretion	Hypoglycemia, weight gain
DPP-4 Inhibitors	Sitagliptin, Linagliptin	$\uparrow$ GLP-1, $\downarrow$ glucagon	Mild weight neutral

SGLT2 Inhibitors	Empagliflozin, Dapagliflozin	↑ Glucose excretion	UTI, DKA, volume depletion
GLP-1 Agonists	Semaglutide, Liraglutide	↓ Appetite, ↓ glucose	GI side effects, weight loss
Thiazolidinediones	Pioglitazone	↑ Insulin sensitivity	Fluid retention, weight gain
Alpha-glucosidase Inhibitors	Acarbose	↓ Carbohydrate absorption	GI symptoms

Insulin Regimens:

- Type 1: Basal-bolus (long-acting + rapid-acting) or insulin pump
- Type 2: Basal insulin (NPH, Glargine, Degludec) + prandial insulin for mealtime coverage

Glycemic Targets (ADA 2026):

- HbA1c < 7% (6.5-7.5% for elderly)
- Fasting: 80-130 mg/dL
- Post-prandial: < 180 mg/dL

3.2 HYPOTHYROIDISM

Definition:

Inadequate production of thyroid hormones, most commonly due to autoimmune (Hashimoto's) thyroiditis.

Symptoms:

- Fatigue, weight gain, cold intolerance
- Constipation, dry skin, hair loss

- Bradycardia, depression, myxedema
- Menstrual irregularities, infertility

Diagnosis:

- ↑ TSH, ↓ Free T4
- Anti-TPO antibodies positive (autoimmune)

Treatment:

- Levothyroxine (T4): Starting dose 1.6 mcg/kg/day
 - Titrate to TSH within normal range (0.5-4.5 mIU/L)
 - For elderly: Start at 25-50 mcg/day, titrate slowly
 - Pregnant: Increase dose by 30-50%, monitor every 4 weeks
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3.3 HYPERLIPIDEMIA

(Refer to Section 1.4 for comprehensive detail)

Summary Protocol:

1. Statin therapy (high-intensity if ASCVD risk > 20%)
 2. Add Ezetimibe if LDL not at goal
 3. PCSK9 inhibitors for familial hyperlipidemia
 4. Lifestyle modifications (Mediterranean diet, exercise)
 5. Target based on risk category (very high risk: LDL < 55 mg/dL)
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CHAPTER 4: NEUROLOGICAL CONDITIONS

4.1 EPILEPSY & SEIZURE DISORDERS

Definition:

Recurrent unprovoked seizures due to abnormal excessive neuronal activity.

Seizure Types:

Type	Characteristics	Examples
Focal (Partial)	Starts in one hemisphere	Simple partial, Complex partial
Generalized	Both hemispheres	Tonic-clonic, Absence, Myoclonic, Atonic
Unknown	Insufficient data	-

Pharmacological Treatment:

A. First-Line for Focal Seizures:

- Lamotrigine, Levetiracetam, Carbamazepine, Oxcarbazepine

B. First-Line for Generalized Tonic-Clonic:

- Valproic acid, Lamotrigine, Topiramate

C. First-Line for Absence Seizures:

- Ethosuximide, Valproic acid, Lamotrigine

D. Broad-Spectrum:

- Valproic acid, Lamotrigine, Levetiracetam

Status Epilepticus Protocol:

1. Benzodiazepines: Lorazepam 0.1 mg/kg IV (max 4 mg) OR Midazolam 0.2 mg/kg
2. Second-line: Fosphenytoin 20 mg/kg IV OR Valproic acid 40 mg/kg IV
3. Third-line: Phenobarbital, Levetiracetam, or continuous infusion (Midazolam, Propofol)
4. If refractory: General anesthesia with Propofol or Thiopental

4.2 STROKE

Definition:

Acute neurological deficit due to vascular injury: ischemia (85%) or hemorrhage (15%).

A. Ischemic Stroke Treatment:

Acute Management (within 4.5 hours):

- Alteplase (tPA): 0.9 mg/kg IV (max 90 mg)
 - 10% bolus, 90% infusion over 60 minutes
 - Contraindications: recent surgery, bleeding, BP > 185/110
- Mechanical Thrombectomy: up to 24 hours (depending on imaging)

Secondary Prevention:

- Antiplatelet: Aspirin 81-325 mg daily or Clopidogrel 75 mg daily
- Dual antiplatelet: Aspirin + Clopidogrel for 21 days (CHANCE trial)
- Statins: High-intensity (Atorvastatin 40-80 mg)
- BP control (target < 130/80)
- Anticoagulation for atrial fibrillation (Warfarin, NOACs)
- Smoking cessation, lifestyle modifications

B. Hemorrhagic Stroke:

- BP control (target SBP < 140 mmHg)
 - Reverse anticoagulation: Vitamin K for Warfarin, Idarucizumab for Dabigatran
 - Surgical evacuation (if cerebellar hemorrhage > 3cm or mass effect)
-

4.3 MIGRAINE

Definition:

Recurrent headache disorder with moderate to severe pain, often unilateral and pulsating.

Symptoms:

- Headache (4-72 hours), unilateral
- Nausea/vomiting, photophobia, phonophobia
- Aura in 25% (visual disturbances, paresthesia)

Acute Treatment:

- Mild: NSAIDs (Ibuprofen 400-600 mg), Acetaminophen
- Moderate: Triptans (Sumatriptan, Rizatriptan, Zolmitriptan)
- Severe: Ditans (Lasmiditan), Gepants (Ubrogepant)
- Intractable: IV NSAIDs + antiemetics (Metoclopramide, Prochlorperazine)

Prophylaxis (if ≥ 4 days/month):

- Beta-blockers: Propranolol, Metoprolol
 - Anticonvulsants: Topiramate, Valproic acid
 - Antidepressants: Amitriptyline, Venlafaxine
 - CGRP Inhibitors: Erenumab, Galcanezumab (monthly injections)
 - Botulinum toxin type A (chronic migraine > 15 days/month)
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CHAPTER 5: GASTROINTESTINAL DISORDERS

5.1 GASTROESOPHAGEAL REFLUX DISEASE (GERD)

Definition:

Retrograde flow of gastric contents into esophagus causing symptoms or mucosal damage.

Treatment:

A. Lifestyle Modifications:

- Elevate head of bed
- Avoid trigger foods (spicy, fatty, acidic, caffeine, alcohol)
- Weight loss if BMI > 25
- Eat smaller meals, avoid eating within 3 hours of bedtime

B. Pharmacological:

- Antacids: For mild infrequent symptoms (Calcium carbonate, Aluminum hydroxide)
- H2 Blockers: Famotidine 20 mg BID, Ranitidine 150 mg BID

- PPIs (First-line): Omeprazole 20 mg daily, Pantoprazole 40 mg daily, Esomeprazole 40 mg daily
 - Duration: 4-8 weeks initial, consider maintenance for chronic cases
 - Risks: Long-term use → B12 deficiency, osteoporosis, C. difficile

C. Surgical:

- Nissen fundoplication (for refractory GERD)

5.2 PEPTIC ULCER DISEASE

Definition:

Erosions in gastric or duodenal mucosa from acid-pepsin secretion imbalance.

Causes:

- H. pylori infection (75% of duodenal ulcers, 60% of gastric)
- NSAID use (risk increases with age, dose, duration)
- Stress, smoking, alcohol

Treatment:

1. H. pylori Eradication (Triple Therapy):
 - Omeprazole 20 mg BID + Amoxicillin 1g BID + Clarithromycin 500 mg BID (14 days)
 - OR Bismuth quadruple: Bismuth subsalicylate + Metronidazole + Tetracycline + PPI (14 days)
2. NSAID-induced ulcers:
 - Stop NSAIDs if possible
 - PPI for 4-8 weeks (healing)
 - Switch to COX-2 selective NSAIDs (Celecoxib) if needed
3. Maintenance:
 - H. pylori eradication confirmed with UBT or stool antigen
 - Long-term PPI for high-risk patients (elderly, history of ulcers)

5.3 IRRITABLE BOWEL SYNDROME (IBS)

Definition:

Functional gastrointestinal disorder with abdominal pain and altered bowel habits.

Types: IBS-D (diarrhea), IBS-C (constipation), IBS-M (mixed)

Treatment:

Symptom	Treatment
Pain	Antispasmodics: Hyoscine, Mebeverine, Peppermint oil
Diarrhea	Loperamide, Rifaximin (for IBS-D), Eluxadoline
Constipation	Fiber, Osmotic laxatives (Miralax), Lubiprostone, Linaclotide
Psychological	SSRIs (for anxiety), Cognitive Behavioral Therapy (CBT)
Microbiome	Probiotics (specific strains: B. infantis, L. plantarum)
Dietary: Low FODMAP diet (avoid fermentable oligosaccharides, disaccharides, monosaccharides, polyols)	

CHAPTER 6: INFECTIOUS DISEASES

6.1 COMMUNITY-ACQUIRED PNEUMONIA (CAP)

(Refer to Section 2.3 for complete detail)

Treatment Duration: 5-7 days (based on clinical response)

Empiric Oral Antibiotics:

- Macrolide: Azithromycin 500 mg x1 → 250 mg daily x4 days
 - Doxycycline 100 mg BID
 - Fluoroquinolone: Levofloxacin 750 mg daily
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6.2 URINARY TRACT INFECTION (UTI)

Definition:

Infection of any part of urinary system. Most common: cystitis, pyelonephritis.

Uncomplicated Cystitis:

- Nitrofurantoin 100 mg BID x 5 days
- OR Trimethoprim/Sulfamethoxazole (TMP/SMX) 160/800 BID x 3 days
- OR Fosfomycin 3g single dose

Acute Pyelonephritis:

- Outpatient: Levofloxacin 750 mg daily x 5-7 days
- Inpatient: IV Ceftriaxone 1-2g daily ± Gentamicin
- Culture-directed therapy (adjust if resistant)

Recurrent UTI Prophylaxis:

- Postcoital dose
 - Methenamine hippurate
 - Low-dose nitrofurantoin 50 mg daily
 - Estrogen cream (postmenopausal)
-

6.3 HIV/AIDS

Definition:

Acquired immunodeficiency syndrome due to Human Immunodeficiency Virus.

Antiretroviral Therapy (ART):

Initial Regimen (First-line):

- Tenofovir/Emtricitabine + (Dolutegravir OR Bictegravir)

Alternative Regimens:

- Abacavir/Lamivudine (if HLA-B*5701 negative) + Dolutegravir
- Tenofovir/Emtricitabine + Efavirenz (generic)

Monitoring:

- CD4 count every 3-6 months
- Viral load every 3 months (goal: < 20 copies/mL)
- HLA-B*5701 testing (prior to Abacavir)
- Renal function (if Tenofovir)
- Lipids and fasting glucose (metabolic complications)

Prophylaxis:

- TMP/SMX for CD4 < 200 cells/ μ L (PCP prophylaxis)
- Azithromycin if CD4 < 50 (MAC prophylaxis)
- Fluconazole if frequent oropharyngeal candidiasis

Pre-exposure Prophylaxis (PrEP):

- Tenofovir/Emtricitabine daily (approved for high-risk individuals)
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CHAPTER 7: AUTOIMMUNE & RHEUMATOLOGIC DISEASES

7.1 RHEUMATOID ARTHRITIS (RA)

Definition:

Chronic systemic autoimmune disease characterized by inflammatory polyarthritis, primarily affecting small joints.

Diagnosis:

- Symmetrical polyarthritis (≥ 6 weeks duration)
- Morning stiffness > 30 minutes
- Rheumatoid factor (RF) and/or anti-CCP positive

- Elevated inflammatory markers (ESR, CRP)
- X-ray: Joint space narrowing, erosions

Pharmacological Treatment:

A. First-line (Disease-Modifying Antirheumatic Drugs - DMARDs):

- Methotrexate (MTX) 15-25 mg/week (with folic acid 1mg daily)
- Sulfasalazine (alternative)
- Hydroxychloroquine (alternative)

B. Biologic DMARDs (if failed MTX):

- TNF- α inhibitors: Etanercept, Adalimumab, Infliximab, Golimumab
- IL-6 inhibitors: Tocilizumab
- T-cell modulation: Abatacept
- B-cell depletion: Rituximab
- JAK inhibitors: Tofacitinib (oral)

C. Steroids:

- Short-term (≤ 3 months) for bridging
- Prednisone 5-10 mg daily (taper)

Non-Pharmacological:

- Physical therapy (range of motion, strengthening)
- Occupational therapy (splints, joint protection)
- NSAIDs for pain

7.2 SYSTEMIC LUPUS ERYTHEMATOSUS (SLE)

Definition:

Autoimmune disease affecting multiple organs, characterized by autoantibody production.

Criteria (SLICC 2012):

- Malar rash, discoid rash, photosensitivity, oral ulcers

- Non-erosive arthritis
- Serositis (pleuritis/pericarditis)
- Renal disorder (proteinuria, cellular casts)
- Neurologic (seizures, psychosis)
- Hematologic (hemolytic anemia, leukopenia, thrombocytopenia)
- Immunologic (anti-dsDNA, anti-Smith, antiphospholipid)
- ANA positive

Treatment:

Severity	Treatment
Mild	Hydroxychloroquine 200-400 mg daily, NSAIDs
Moderate	Glucocorticoids (Prednisone 10-20 mg daily), immunosuppressants
Severe (organ-threatening)	High-dose steroids + Cyclophosphamide or Mycophenolate mofetil
Refractory	Rituximab, Belimumab

Monitoring:

- Renal function, urinalysis (proteinuria)
- Complement levels (C3, C4)
- Anti-dsDNA titers
- Annual ophthalmology exam (due to hydroxychloroquine toxicity)

CHAPTER 8: PSYCHIATRIC & NEUROLOGICAL DISORDERS

8.1 MAJOR DEPRESSIVE DISORDER (MDD)

Definition:

Persistent depressed mood, anhedonia, with functional impairment ≥ 2 weeks.

Pharmacological Treatment:

Class	Examples	Starting Dose	Side Effects
SSRIs	Fluoxetine, Sertraline, Escitalopram	20 mg, 50 mg, 10 mg	GI, sexual dysfunction, activation
SNRIs	Venlafaxine, Duloxetine, Desvenlafaxine	75 mg, 30 mg, 50 mg	Hypertension, sweating
Tricyclics	Amitriptyline, Nortriptyline	25-50 mg, 10-25 mg	Anticholinergic, cardiotoxic
Atypicals	Bupropion, Mirtazapine, Vortioxetine	150 mg, 15 mg, 10 mg	Bupropion lowers seizure threshold

Treatment Algorithm:

1. SSRI/SNRI/Bupropion for 6-8 weeks
2. Increase dose or switch if partial response
3. Augmentation with second antidepressant, mood stabilizer, or atypical antipsychotic
4. ECT for treatment-resistant or severe depression

Psychotherapy: CBT, IPT, psychodynamic therapy

8.2 ANXIETY DISORDERS

Generalized Anxiety Disorder (GAD):

- SSRIs/SNRIs (first-line): Sertraline, Escitalopram, Venlafaxine
- Buspirone (non-sedating)
- Benzodiazepines (limited, short-term): Alprazolam, Lorazepam, Clonazepam

Panic Disorder:

- SSRIs/SNRIs (first-line)
- CBT (essential)
- Benzodiazepines (for acute attacks only)

Social Anxiety Disorder:

- SSRIs/SNRIs
 - Beta-blockers (propranolol for performance anxiety)
 - CBT
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CHAPTER 9: ONCOLOGY & HEMATOLOGY

9.1 BREAST CANCER

Definition:

Malignancy of breast epithelial tissue (ductal 75%, lobular 10%).

Treatment by Stage:

Stage	Treatment
0 (DCIS)	Lumpectomy + radiation OR mastectomy

I-II	Surgery + chemotherapy, HER2-targeted therapy, endocrine therapy
III	Neoadjuvant chemotherapy + surgery + radiation + adjuvant
IV	Systemic therapy (palliative): chemotherapy, hormonal therapy, targeted

Targeted Therapy:

- HER2-positive: Trastuzumab (Herceptin), Pertuzumab, Ado-trastuzumab emtansine
- Hormone receptor-positive: Tamoxifen, Aromatase inhibitors (Anastrozole, Letrozole)
- CDK4/6 Inhibitors: Palbociclib, Abemaciclib (advanced HR+/HER2-)

Endocrine Therapy:

- Premenopausal: Tamoxifen 20 mg daily x 5-10 years
- Postmenopausal: Anastrozole 1 mg daily or Letrozole 2.5 mg daily x 5-10 years

Surveillance: Mammography annually, clinical exam every 3-6 months for 3 years

9.2 ANEMIA

Definition:

Reduced RBC mass, hemoglobin below normal for age/sex.

Type	Diagnosis	Treatment
------	-----------	-----------

Iron Deficiency	↓ Ferritin, ↓ Iron, ↑ TIBC	Ferrous sulfate 325 mg TID (with Vit C)
B12 Deficiency	↓ B12, ↑ MCV, ↑ MMA	Cyanocobalamin IM injections
Folate Deficiency	↓ Folate, ↑ MCV	Folic acid 1 mg daily
Anemia of Chronic Disease	↓ Iron, ↓ TIBC, ↑ Ferritin	Treat underlying disease
Aplastic Anemia	Pancytopenia, hypocellular marrow	Immunosuppression, transplant
Hemolytic Anemia	↑ Reticulocytes, ↑ LDH, ↓ Haptoglobin	Treat cause; steroids

Hemoglobin Targets:

- Men: > 13 g/dL
- Women: > 12 g/dL
- Pregnant: > 11 g/dL

9.3 LEUKEMIA

Definition:

Malignant neoplasm of hematopoietic tissue.

Type	Characteristics	Treatment
------	-----------------	-----------

AML	Myeloid blasts > 20%	Chemotherapy (7+3), stem cell transplant
ALL	Lymphoid blasts; children	Vincristine, Prednisone, Asparaginase
CML	Philadelphia chromosome t(9;22)	Tyrosine kinase inhibitors (Imatinib)
CLL	B-cell proliferation; older adults	BTK inhibitors (Ibrutinib, Zanubrutinib), venetoclax, rituximab

CHAPTER 10: DERMATOLOGICAL CONDITIONS

10.1 ACNE

Definition:

Disorder of pilosebaceous units causing comedones, papules, pustules, nodules.

Treatment Algorithm:

- Mild: Topical benzoyl peroxide, salicylic acid, retinoids (Adapalene)
- Moderate: Combination (Benzoyl peroxide + Clindamycin) OR (Adapalene + Benzoyl peroxide)
- Severe: Oral antibiotics (Doxycycline, Minocycline, Sarecycline) + Topical retinoids
- Severe (Refractory): Isotretinoin 0.5-1 mg/kg/day x 4-6 months (requires monitoring for teratogenicity, lipids, mood changes)

Hormonal Therapy (Females):

- Spironolactone (anti-androgen)

- Combined oral contraceptives (Yasmin, Ortho Tri-Cyclen)
-

10.2 PSORIASIS

Definition:

Chronic immune-mediated disorder with T-cell activation, hyperproliferation of keratinocytes.

Treatment:

Severity	Treatment
Mild (< 5% BSA)	Topical corticosteroids, Vitamin D analogs (Calcipotriene), Salicylic acid
Moderate (5-10% BSA)	Phototherapy (UVB, PUVA), Systemic agents (Methotrexate, Cyclosporine)
Severe (> 10% BSA)	Biologics: TNF- α (Etanercept, Adalimumab, Infliximab), IL-17 (Secukinumab, Ixekizumab), IL-23 (Ustekinumab, Guselkumab)

Monitoring:

- Liver function (Methotrexate)
 - Renal function (Cyclosporine)
 - TB screen prior to biologics
-

10.3 ECZEMA (Atopic Dermatitis)

Definition:
Chronic pruritic inflammatory skin disease with genetic predisposition.

Treatment:

- Emollients: (First-line, daily hydration)
- Topical corticosteroids: Mild (Hydrocortisone) to potent (Clobetasol)
- Topical calcineurin inhibitors: Tacrolimus, Pimecrolimus (for sensitive areas)
- Systemic: Oral corticosteroids (short-term), Cyclosporine, Methotrexate
- Biologics: Dupilumab (anti-IL-4R α) - for moderate to severe refractory cases
- Antihistamines: Cetirizine, Loratadine (for pruritus)

CHAPTER 11: RENAL & UROLOGICAL DISEASES

11.1 CHRONIC KIDNEY DISEASE (CKD)

Definition:
Progressive loss of kidney function for ≥ 3 months.

Staging:

Stage	GFR (mL/min/1.73m ²)	Management
1	≥ 90	Diagnosis, treat comorbid conditions
2	60-89	Slow progression, monitor
3a	45-59	Evaluate complications
3b	30-44	Manage anemia, bone disease

4	15-29	Prepare for renal replacement
5	< 15 or dialysis	Replacement therapy

Treatment:

- BP target: < 130/80 (ACE inhibitors/ARBs are preferred)
- Diabetic control: HbA1c < 7%
- Lipid control: Statin
- Anemia: Erythropoietin (Epoetin alfa) if Hb < 10 g/dL
- Bone disease: Vitamin D, phosphate binders
- Diet: Protein restriction (0.6-0.8 g/kg/day), sodium < 2g, potassium and phosphorus restriction

11.2 NEPHROLITHIASIS (Kidney Stones)

Definition:

Calculi formation in kidney from crystallized supersaturated substances.

Type	Composition	Treatment
Calcium oxalate (80%)	Most common	Thiazides + Potassium citrate
Calcium phosphate (10%)	-	Acetazolamide, restrict calcium
Uric acid (5-10%)	Gout, high purine	Allopurinol, alkalize urine
Struvite (infections)	Magnesium ammonium phosphate	Antibiotics, surgery

Medical Management:

- Pain: NSAIDs (Ketorolac), IV opioids (for large stones)
 - Small stones (< 5 mm): Conservative (hydration, strain urine, α -blockers to facilitate passage)
 - Stones > 5-10 mm: ESWL, Ureteroscopy, or PCNL
-

CHAPTER 12: MUSCULOSKELETAL DISORDERS

12.1 OSTEOARTHRITIS (OA)

Definition:

Degenerative joint disease (non-inflammatory) affecting weight-bearing joints.

Non-Pharmacological:

- Weight loss, exercise (aquatic, low-impact), physical therapy
- Assistive devices (canes, braces)

Pharmacological:

Category	Examples	Notes
Acetaminophen	325-650 mg Q6H	First-line; hepatotoxic at high doses
NSAIDs	Ibuprofen, Naproxen, Celecoxib	Effective; GI risk, renal risk
Topical NSAIDs	Diclofenac gel	Good for localized pain
Topical Capsaicin	Capsaicin cream	For refractory cases

Corticosteroid injections	Triamcinolone	3-4 months relief
Hyaluronic acid	Viscosupplementation	Knee OA; variable efficacy
Duloxetine	SNRI	For chronic pain (back pain)

12.2 OSTEOPOROSIS

Definition:

Low bone mass with increased fracture risk (T-score $\leq$ -2.5).

Treatment:

Class	Examples	Indication	Notes
Bisphosphonates	Alendronate, Zoledronic acid, Risedronate	First-line	GI intolerance, on therapy for 3-5 years
SERMs	Raloxifene	Postmenopausal osteoporosis	Reduce breast cancer risk
RANKL Inhibitor	Denosumab	High fracture risk	Twice-yearly injection
PTH Analogs	Teriparatide, Abaloparatide	Severe osteoporosis	Anabolic therapy
Sclerostin Inhibitor	Romosozumab	Postmenopausal with high risk	Monthly injection, 12 months

Prevention:

- Calcium 1000-1200 mg/day (dietary + supplement)
- Vitamin D 800-2000 IU/day
- Weight-bearing exercise
- Fall prevention

APPENDIX A: DRUG INTERACTIONS & CONTRAINDICATIONS

A.1 Important Drug Interactions

Drug	Interacting Drug	Effect	Management
Warfarin	Broad-spectrum antibiotics	↑ INR → bleeding risk	Monitor INR closely
Statins	Fibrates	↑ Myopathy risk	Avoid combination
ACE Inhibitors	Potassium-sparing diuretics	↑ Hyperkalemia	Monitor potassium
Digoxin	Verapamil, Amiodarone	↑ Digoxin levels	Reduce digoxin dose
MAO Inhibitors	SSRIs	Serotonin syndrome	Avoid combination
Lithium	NSAIDs	↑ Lithium toxicity	Monitor lithium levels

A.2 Common Contraindications

Drug	Contraindication	Alternative
ACE Inhibitors	Pregnancy	ARBs
Metformin	eGFR < 30 mL/min	Sulfonylureas, Insulin
Aspirin	Active GI bleed	Clopidogrel
Tetracyclines	Pregnancy, children < 8	Macrolides
Statins	Active liver disease	Ezetimibe
Isotretinoin	Pregnancy (teratogenic)	Alternative acne treatment

APPENDIX B: LABORATORY REFERENCE RANGES

Test	Reference Range	Critical Values
Hemoglobin (M)	13.5-17.5 g/dL	< 7 g/dL
Hemoglobin (F)	12.0-16.0 g/dL	< 7 g/dL
Hematocrit (M)	38-52%	< 20%
Hematocrit (F)	35-47%	< 20%
WBC	4,500-11,000/ μ L	< 2,000 or > 30,000

Platelets	150,000-400,000/ μ L	< 20,000
Glucose (Fasting)	70-100 mg/dL	< 40 or > 400
HbA1c	< 5.7%	-
BUN	7-20 mg/dL	> 100 mg/dL
Creatinine	0.6-1.2 mg/dL	> 4 mg/dL
Sodium	135-145 mEq/L	< 120 or > 160
Potassium	3.5-5.0 mEq/L	< 2.5 or > 6.5
TSH	0.5-4.5 mIU/L	> 100 mIU/L
LDL	< 100 mg/dL (ideal)	-
HDL (M)	> 40 mg/dL	-
HDL (F)	> 50 mg/dL	-
Total Cholesterol	< 200 mg/dL	-
CRP	< 5 mg/L	> 50 mg/L
ESR (M)	0-15 mm/hr	> 100 mm/hr
ESR (F)	0-20 mm/hr	> 100 mm/hr
Ferritin (M)	20-500 ng/mL	-

Ferritin (F)

20-200 ng/mL

-

APPENDIX C: EMERGENCY PROTOCOLS

C.1 Anaphylaxis Protocol

1. First-line: Epinephrine 0.3-0.5 mg IM (1:1000) lateral thigh
2. Airway management: High-flow oxygen, consider intubation
3. IV fluids: Normal saline bolus (1-2L)
4. Anti-histamines: Diphenhydramine 50 mg IV
5. Corticosteroids: Methylprednisolone 125 mg IV (prevent biphasic reaction)
6. Bronchodilators: Albuterol if bronchospasm present
7. Monitoring: Vital signs every 5-15 minutes for at least 4-6 hours

C.2 Acute Myocardial Infarction Protocol

1. MONA: Morphine, Oxygen (if SpO₂ < 90%), Nitroglycerin, Aspirin
2. Anti-platelet: Aspirin 325 mg chewable + Ticagrelor 180 mg
3. Anticoagulation: Heparin bolus + infusion
4. Beta-blocker: Metoprolol 5 mg IV (if no heart failure, low BP)
5. Statins: High-intensity atorvastatin 80 mg
6. Reperfusion: PCI within 90 minutes; If not feasible, fibrinolytic within 30 minutes

C.3 Stroke (Ischemic) Protocol

1. NIHSS assessment (within 10 minutes)
2. CT Head without contrast (within 25 minutes)
3. Check blood glucose (correct if < 60 or > 400)
4. Alteplase (tPA) within 4.5 hours (0.9 mg/kg; 10% bolus, 90% over 60 min)
5. BP control: If tPA given, SBP < 185, DBP < 110
6. Notify thrombectomy team if large vessel occlusion

C.4 Status Epilepticus Protocol

1. 0-5 minutes: Lorazepam 0.1 mg/kg IV (max 4 mg) OR Midazolam IM (10 mg if > 40 kg)
 2. 5-20 minutes: Fosphenytoin 20 mg/kg IV OR Valproic acid 40 mg/kg IV
 3. 20-40 minutes: Levetiracetam 40 mg/kg IV
 4. > 40 minutes (Refractory): Propofol OR Midazolam infusion (intubation, ICU admission)
 5. Check: Glucose, electrolytes, anticonvulsant levels, EEG
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